

TO DO

- ☒ ~~More species~~
- ☐ Detection task benchmark
 - ☐ Can use Birdset labels
- ☒ ~~Compelling figure needed to show what the granularity of tokenization means~~
- ☒ ~~MSE error of held-out species~~
- ☐ Host Mirror (zenodo) + annotations for our benchmarks
 - ☐ Declare this a new benchmark -- justify that nothing standard really exists
- ☐ That one beyer paper talks about how scaling MAE / VIT can be thought about patch granularity instead of parameters only
 - ☐ Show Tim the interactions with model patches / masking and C parameter
- ☐ Say patch design is a KEY inductive bias for song encoder
- ☐ **We also would like to note that SongMAE beats HuBERT and BirdAVES despite its lower dim, which makes linear probes and MLP finetunes much harder + much lower parameters**
- ☐ **Read old interspeech preprint to see if i missed some citations or sentences that are cool**
- ☐ **Vector-db , cosine similarity between species**
- ☐ **Benefit is it can leverage all the segmentation methods developed for ViT's**
- ☐ **Mention how we use the residual stream for UMAP (we cite several anthropic papers for this)**
- ☐ **Reiterate and cast doubt on the fact of canary,zf,bf being in the BirdAVES train data**

SongMAE: A bioacoustic encoder for birdsong

George Vengrovski, Tim Gardner

Abstract

The architecture of existing self-supervised bioacoustic encoders has largely been inherited from human speech models, and as a result these encoders operate at coarse temporal scales. This is well suited to the canonical bioacoustic tasks of species classification and song detection, but it lacks the temporal resolution needed to distinguish the individual syllables and notes that compose birdsong. We introduce SongMAE, a masked autoencoder (MAE) pretrained on spectrograms of birdsong. Unlike typical audio MAEs, which divide spectrograms into square patches spanning equal extents in frequency and time, SongMAE uses patches just 5 ms wide in time and decouples the frequency extent of a patch from its temporal extent, so the tokenization granularity can be set independently along each axis. This flexibility is enhanced by Voronoi-based spatial masking, which produces irregular, connected masked regions that prevent the trivial reconstruction that fine temporal patches would otherwise permit through local interpolation. We evaluate several patch configurations across various tasks. We find that SongMAE outperforms existing bioacoustic encoders for syllable classification tasks, reconstructs masked spectrograms of species absent from its pretraining data, yields latent spaces where syllable types cluster without supervision, and when finetuned with limited labels it achieves strong frame-level syllable classification, matching or exceeding a model with an order of magnitude more parameters.

Index Terms: bioacoustics, birdsong, self-supervised learning, masked autoencoders, syllable classification

1. Introduction / Motivation

Bioacoustics has seen a proliferation of pretrained self-supervised encoding models. Models such as BirdMAE [1], AVES [2], and BEAT [3] have been pretrained on a large corpus of bioacoustic data, motivated by species classification and detection in data-sparse contexts. These models operate on multi-second windows (typically 5 s), classifying which species is singing or detecting whether a song is present. Though state-of-the-art for such temporally coarse tasks, they were not designed to represent the fine-grained components of song notes, syllables, or calls, which transition within milliseconds and last only tens of milliseconds. Tasks like classifying syllable types at precise temporal locations demand exactly this granularity, and creating useful embeddings for these elements remains an open problem requiring models specialized for rapid temporal structure.

Existing bioacoustic embedding models typically segment audio into tokens—either via 1D convolution of raw samples (WhisperSeg, AVES) [2, 4] or, in ViT-like models, by dividing spectrograms into patch embeddings [1]. This segmentation introduces a fundamental trade-off in temporal resolution. Species classification benefits from coarse resolution, which reduces memory use given transformers’ quadratic attention scaling, but coarse tokens or patches span temporal spans that exceed the shortest syllables and intersyllabic gaps in birdsong. These models are therefore inadequate for song structure analysis, which requires fine-grained resolution to capture the rapid acoustic transitions defining individual vocalizations.

No general bioacoustic model optimizes for this resolution. TweetyBERT [5] demonstrates the value of high temporal resolution for syllable discovery but remains canary-specific. Its masking strategy is also constrained to a single spatial configuration—masks spanning all frequencies within specific time blocks—and whether this constraint benefits or limits generalization to other species is an open question we revisit in our ablation study (Section 4.3).

To address this gap, we introduce SongMAE, a Masked Autoencoder (MAE) vision transformer (ViT) pretrained on a portion of the BirdSet dataset [6]. Our goal is to test whether a general syllable-level model can learn representations that generalize to species and recording environments outside its training set. We have specifically taken steps to exclude species of birdsong for which we have high-quality annotations, such as the canary, from the pretraining corpus, so that we can analyze how our model performs on them. Our key technical innovation lies in patch design: we compute spectrograms at 5 ms time-bin resolution and employ asymmetric patches spanning multiple mel bins but only a single time bin, preserving fine temporal granularity. We combine this patching with a Voronoi-based masking approach, which provides a modest boost. We evaluate patch configurations of 16, 32, and 128 mel bins per patch, all maintaining 5 ms temporal width, and find that performance is robust across these settings. To evaluate our architectural setup independently of temporal resolution, we also train SongMAE at the same temporal resolution as BirdAVES, an existing bioacoustic encoder, allowing a direct comparison under matched output granularity.

We find that SongMAE is strong at syllable-level analysis of birdsong. Its learned representations transfer across every downstream setting we test: reconstruction of spectrograms, linear probes, unsupervised syllable clustering, and fine tuning under limited labels, all of which were done with songbird species omitted during pretraining. We also find that the patch shape need not be square, as in canonical MAEs. Decreasing frequency resolution has a diminishing effect on performance while carrying a substantial computational cost, so compute is better spent on fine temporal resolution, which reduces errors in classifying the rapid switching dynamics of birdsong. Although the SongMAE models we evaluate are likely underparameterized and undertrained, they already show promise for being employed in bioacoustic tasks, and likely will only improve with further scaling.

2. Related Work

Masked autoencoders (MAEs) have emerged as a powerful self-supervised pretraining framework for audio [7, 8]. Adapted from their success in vision, audio MAEs operate on spectrogram representations: the input spectrogram is divided into patches, a large fraction of patches are randomly masked, and an encoder-decoder transformer learns to reconstruct the missing regions. By forcing the model to predict masked content from partial context, MAEs learn rich representations without requiring labeled data. The resulting encoder can then be finetuned or used as a feature extractor for downstream tasks.

Prior work on audio MAEs shows that finer temporal patching can improve downstream performance. In MSM-MAE, Niizumi et al. found that reducing patch width over time (16 mel bins $\times$ 8 time bins and 16 mel bins $\times$ 4 time bins vs. 16 mel bins $\times$ 16 time bins) improved performance across most evaluation tasks, with the strongest gains observed in speech and music benchmarks [9]. In MAE-AST, Baade et al. further showed that for speech classification, frame-based tokenization spanning the full mel axis (128 mel bins $\times$ 2 time bins) outperformed standard patch-based

tokenization (16 mel bins $\times$ 16 time bins), although patch-based inputs remained stronger for some audio-event tasks [10]. Consistent with this broader pattern, Wu et al. showed in wav2vec 2.0 efficiency-performance analyses that reducing temporal resolution degrades performance at equivalent model capacity [11].

A preliminary study focused exclusively on canary appeared as a non-archival workshop paper [12]; the present study extends it to a multi-species evaluation with analysis of latent spaces, ablations, and supervised fine-tuning experiments.

3. General Methods

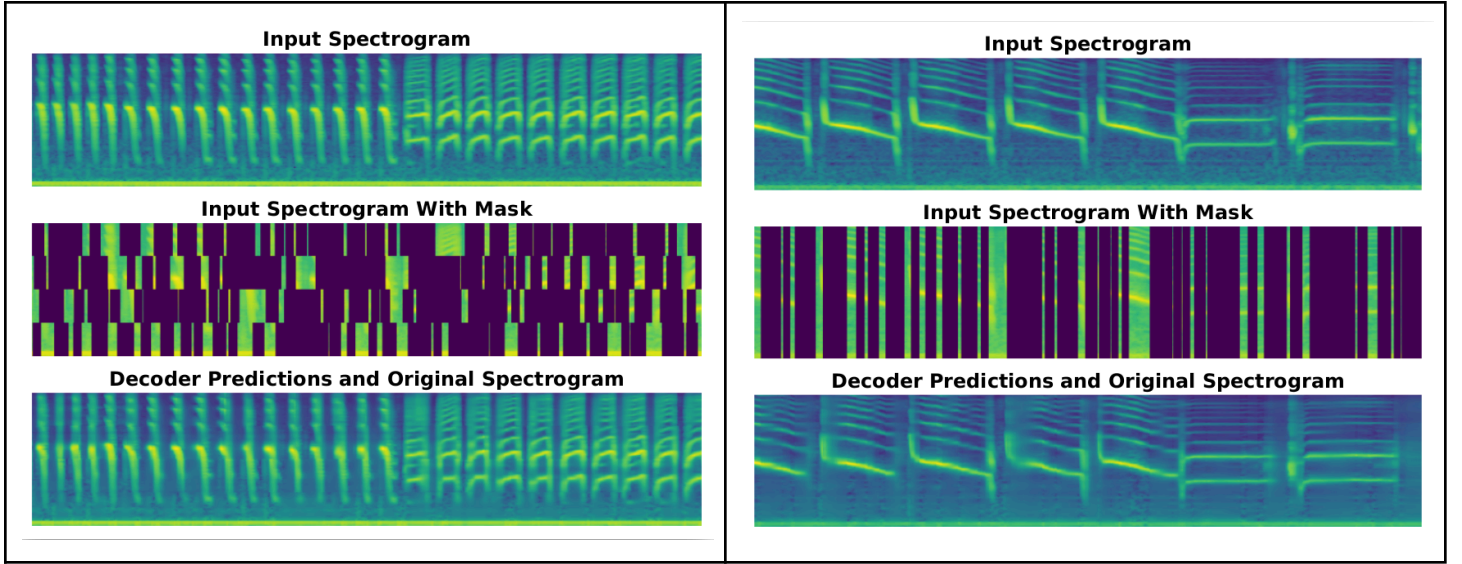


Figure 1: Two examples of SongMAE reconstructions of canary song (unseen species) during training: the 1×32 (mel $\times$ time-bin) model (left) and the 1×128 model (right). Top: canary spectrogram. Middle: masks applied prior to the forward pass. Bottom: decoder output overlaid on the original spectrogram.

3.1. Architectural Overview

SongMAE follows the MAE architecture: an asymmetric encoder-decoder operating on spectrograms [1, 7, 8]. Spectrograms are divided into patches via a convolutional projection, positionally encoded, and then masked at a high ratio. Before the encoder, we apply a shallow stack of convolutional layers over the full patch grid, with mask tokens substituted at the masked positions. These layers add local inductive bias that the bioacoustic domain particularly benefits from: birdsong recordings are noisy, and convolution helps the model aggregate local time-frequency context and suppress this noise rather than learning these local correlations from scratch through attention alone. Convolution is also more sample efficient than self-attention, a property that hybrid convolution-attention architectures such as the Conformer [REF] exploit to strong effect. We do not adopt a full Conformer, however, because interleaving convolution throughout the encoder would require passing masked patches to the encoder, which is incompatible with the MAE design we rely on, so we confine convolution to a single block before the encoder and encode only the visible patches, whose representations are then combined with learned mask tokens and passed to the decoder.

Before decoding, we apply a second convolutional block over the reassembled full patch grid, now containing both encoded visible tokens and mask tokens, so that the decoder also benefits from local aggregation when reconstructing the masked regions. The decoder then reconstructs the masked spectrogram patches using mean squared error (MSE) loss.

3.2. Patch Shapes

The key innovation in SongMAE is the use of patches whose shape is optimized for fine temporal resolution. MAE almost always uses square patches, meaning a patch is generated from an equal number of pixels in both dimensions, and MAEs applied to the bioacoustic domain follow this approach: they employ square patches (e.g. 16 mels $\times$ 16 time bins) spanning hundreds of milliseconds per patch, a design well-suited to the species classification they target [1, 8]. For spectrograms, however, the two axes are not equivariant the way they are for images, that is, a given number of pixels along the frequency axis does not carry the same quantity of information as the same number along the temporal axis. We therefore explore a wide range of patch shapes, comparing rectangular patches that span many mel bins but only a single time bin against typical square patches. The key strength of rectangular patches is that they achieve substantially finer temporal granularity, letting the model resolve the rapid syllable transitions that fall below the resolution floor of existing encoders while preserving memory.

Given our 128-dimensional mel spectrograms, we evaluate patch heights of 16, 32, and 128 mel bins, all maintaining 5 ms temporal width, and find performance robust across these configurations (Section 4.3), we also compare these shapes to the standard square shapes 4 mel $\times$ 4 timebins.

3.3 Voronoi Masking

Fine-grained temporal-frequency patches pose a challenge for masking: standard random masking allows trivial reconstruction through local correlations if the patch spans only a few ms in width. We address this with Voronoi-based spatial masking. Seed patches are initialized according to parameter C (set to 0.1), which represents the fraction of patches that serve as seeds. Each non-seed patch is assigned the Euclidean distance to its nearest seed, accounting for patch dimensions, and the 75% of non-seed patches with the lowest distances are masked. Higher values of C produce many small masked regions, while lower values produce fewer, larger contiguous regions; in both cases, the resulting masks are irregular and connected, forcing the model to learn both local and global dependencies rather than relying on simple interpolation. The approach is agnostic to patch dimensions and requires only a single interpretable parameter (C) to control mask geometry.

We use a 75% masking ratio, consistent with the standard MAE framework, which has been shown to work well across vision and audio domains [1, 7], and search over several C values.

3.3. Pretraining and Test Datasets

The pretraining dataset consists of the XCL subset of BirdSet [6], derived from Xeno-Canto recordings [13], totaling 528,434 recordings. We split this into 95% training and 5% test to monitor overfitting. After the split, we search for the birds that we evaluate the model on, and move them out of the training set.

For evaluation, we use laboratory-recorded datasets with high-quality ground-truth syllable-level labels for canary [14] (3 birds; per bird: $\sim 27 \pm 5$ syllable classes, $\sim 493 \pm 159$ mins), Bengalese finch [15] (11 birds; per bird: $\sim 9 \pm 3$ syllable classes, $\sim 44 \pm 22$ mins), and zebra finch [16] (14 birds; per bird: $\sim 9 \pm 2$ syllable classes, $\sim 5 \pm 2$ mins), totaling 8,014 recordings. Crucially, none of these species or datasets were present in pretraining.

3.4. Spectrogram Generation

Audio files are resampled to 32 kHz and converted to 128 dimensional log-mel spectrograms. We use an FFT size of 1024 with a Hanning window and hop length of 64 samples, producing time bins of 5 ms. The spectrograms are normalized using dataset-wide z-scoring statistics (mean: -50.14, std: 20.66).

3.5. Pretraining Details

We train for 100,000 steps with a 75% masking ratio, using the AdamW optimizer with a learning rate of 1×10^{-4} , weight decay of 0.01, 5,000 warmup steps, and cosine annealing that decays to 1×10^{-5} [17, 18]. Reconstruction loss follows the standard MAE approach: patch-wise MSE computed only on masked patches. We use a batch size of 128 with automatic mixed precision for training speed and memory efficiency [19].

We test three model sizes: micro, tiny, and base (Table X). Micro is used for the ablation sweeps, where many full pretraining runs make a small model necessary, and base produces our final reported numbers. Tiny sits between them as an intermediate jump, letting us see the effect of scaling. Even the base configuration is small by modern neural network standards; we were limited by compute and pretrain data, and by context length, since the fine temporal patches produce long token sequences that make the quadratic attention cost the binding constraint -- because of this, we limited our investigation to smaller models.

	Micro	Tiny	Base
Encoder_D	128	256	384
Encoder Heads	2	4	6
Encoder Layers	6	6	6
Decoder_D	64	128	192
Decoder heads	1	2	3
Parameters (millions)	1.75 mil	6.72 mil	14.89 mil

Table 1: *Architecture and parameter counts for the three SongMAE model sizes. Encoder depth is held fixed at 6 layers across sizes; width and decoder capacity scale with size. The feed-forward hidden dimension is $3 \times$ the attention dimension in every block.*

3.6. Supervised Training Details

For both the linear probe and the MLP-head experiments, we adopt a simple classification head following the approach shown to be effective in AVES [2]: a head producing one prediction per time step. For patch configurations where multiple patches tile the frequency axis (e.g 16 mels $\times$ 1 time

bin), the encoder produces multiple patch embeddings per time step, which we concatenate into a single vector before passing to the head. For the 128 mels $\times$ 1 time bin configuration and for AVES, which both produce one embedding per time step, no concatenation is needed. Since ground-truth labels exist at finer temporal resolution than the patch grid across all experiments, we downsample labels to match the model's temporal patch width.

We evaluate two supervision regimes, each targeting a different question. The first is a linear probe (encoder frozen, single linear layer) used to assess the quality of the learned representations: if syllable-relevant structure is linearly decodable, the pretraining objective has already organized the embedding space in a useful way. The second is a frozen-encoder MLP head, which we use to evaluate performance in data-sparse settings. We adopt this rather than fine tuning the full encoder because the realistic deployment scenario provides only seconds of labeled audio per bird, far too little to update the full set of encoder weights without severe overfitting; training a small head on top of fixed representations is the appropriate regime under such label budgets.

For each bird, we split the songs into a train and test pool, where 20 percent of the songs go into the test pool. We ensure that each syllable is present in both test and train. For the frozen-encoder MLP-head experiments, we simulate limited-label settings by adding whole songs up to a duration budget, covering every class first and then filling the budget with the longest songs. The linear probe is logistic regression trained up to 1,000 iterations. The frozen-encoder MLP head is trained with AdamW and cross-entropy loss at a constant learning rate of 1e-3, using a batch size of 4,096 for 20 epochs.

3.7. Evaluation Metrics

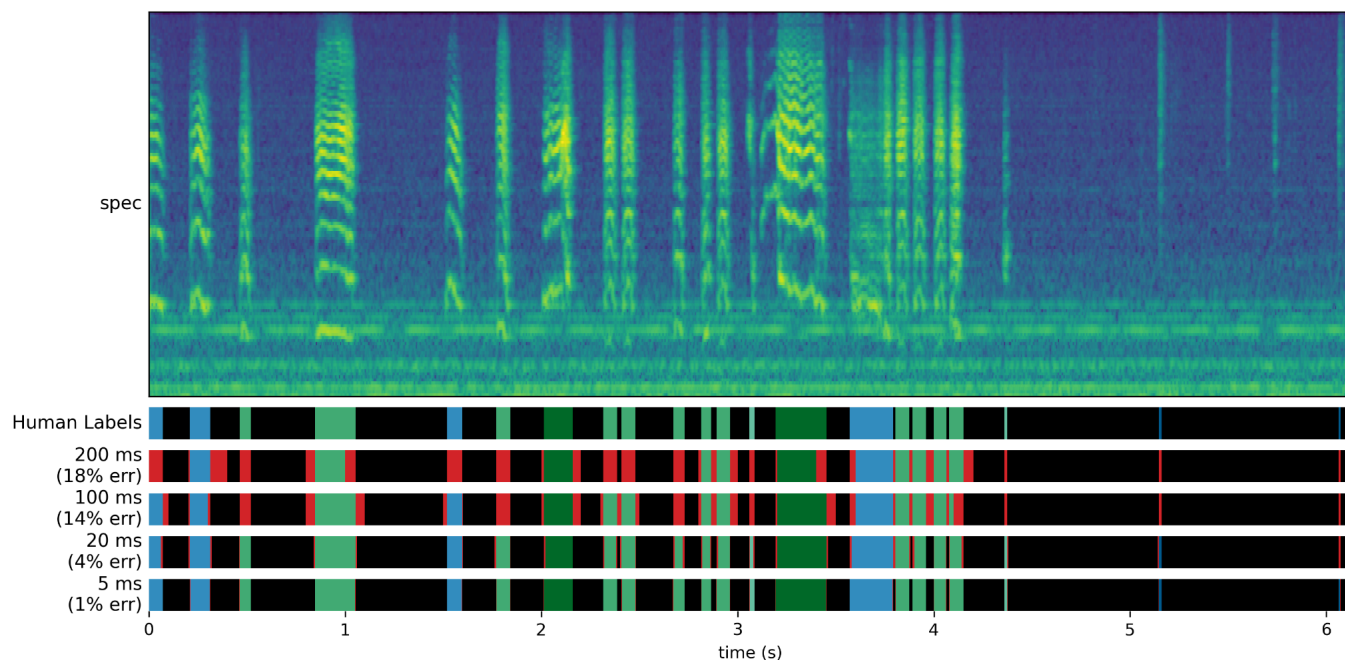


Fig 2 | A zebra finch spectrogram, with the best possible predictions for each level of granularity of tokenization. Tokenization that is above 20ms has difficulty representing silences between syllable types, as well as very short acoustic events. 20ms occasionally straddles silence classes and short acoustic clicks.

We evaluate using a single metric: macro frame error rate (Macro FER). FER is an established metric for the analysis of birdsong [5, 14], providing an interpretable per-frame misclassification rate. However, because syllable repertoires are highly imbalanced—in complex singers like canaries, some syllable types occur an order of magnitude less frequently than common ones—standard FER makes it possible to achieve high scores by simply predicting common syllables. Detecting these rare syllables is critical for characterizing song syntax, individual identity, and repertoire completeness. We therefore report Macro FER, averaging the per-frame accuracies of each syllable type so that rare and common syllables contribute equally.

A methodological challenge arises from temporal resolution differences between models. Ground-truth annotations provide continuous onset and offset times for each syllable along with its class label, but all models output discrete predictions corresponding to fixed temporal spans (tokens or patches). To enable fair comparison, we adopt the following procedure: ground-truth boundaries are rounded to the nearest millisecond, each model’s discrete predictions are expanded into a sequence of 1 ms frames (assigning all frames within a prediction span the same label), macro FER are computed over these aligned sequences. This upsampling step is essential—without it, models with coarse predictions would not be penalized for straddling syllable boundaries.

This procedure reveals an inherent advantage for models with finer temporal resolution. Even if two models learn equally strong internal representations, the model with coarser output resolution will produce predictions that straddle syllable boundaries or miss brief syllables entirely, resulting in lower scores. We consider this appropriate rather than a confound: accurate labeling of short-duration syllables, brief inter-syllabic silences, and rapid transitions between vocal elements is precisely the capability required for fine-grained song analysis.

4. Model Ablations and Experiments

For all analysis, we compare SongMAE against two encoders: BirdAVES (AVES) and HuBERT. BirdAVES is a variant of the AVES encoder that was further pretrained on bioacoustic data to improve performance on bird vocalization tasks, making it the strongest available baseline for this comparison. BirdAVES is a HuBERT-based self-supervised encoder (~316M parameters) pretrained on Xeno-Canto recordings at 16 kHz with 20 ms frame-level embeddings [2]. We also compare against the original HuBERT, which serves as a control for human speech models and, because it shares BirdAVES's architecture, isolates the contribution of bioacoustic pretraining from architecture alone. We exclude models operating at resolutions too coarse for syllable-level tasks (e.g., BirdMAE, 160 ms patches), supervised species-specific networks (TweetyNET, DAS) [14, 24], and models that are designed to be finetuned for bioacoustic tasks WhisperSeg [4].

To identify the optimal configuration of SongMAE, we search over both masking and patch shape parameters. All ablations use the pretraining settings described in Section 3.3 and the micro model size to keep the sweep tractable. Because each configuration requires a full pretraining run, an exhaustive grid over patch and masking parameters is computationally prohibitive, so we structure the search to isolate one factor at a time. We first probe masking in two targeted ablations: a small C interaction ablation crossing the 32×1 over C in $\{0.025, 0.05, 0.1, 0.2\}$, followed by a masking-strategy comparison (32×1 random vs Voronoi). Carrying forward the best C, we then pretrain five tokenization variants under matched settings: 128×1 , 32×1 , 16×1 , 32×4 , and a 4×4 square-patch control (mel bins $\times$ time bins). These support four clean comparisons: 32×1 vs 32×4 isolates temporal resolution, 128×1 vs 32×1 vs 16×1 isolates frequency patch height, 32×4 vs AVES/BirdAVES matches the 20 ms output rate of existing encoders, and 4×4 vs 32×4 contrasts square against asymmetric patch shapes.

5.1. Linear Probe Performance

Linear Probe Ablations	-	-	-	-
Config	Canary	Zebra	Bengalese	Mean
Voronoi C parameter (32x1)	-	-	-	-
C=0.025	8.13	15.59	2.67	8.8
C=0.05	5.63	13.3	2.35	7.09
C=0.1	5.94	13.48	2.27	7.23
C=0.2	6.19	13.51	2.23	7.31
Masking (32x1)	-	-	-	-
Random	6.6	13.74	2.42	7.59
Voronoi	5.63	13.3	2.35	7.09
Patch shape (Voronoi, C=0.1)	-	-	-	-
128x1	13.72	18.19	3.76	11.89
32x1	5.63	13.3	2.35	7.09
16x1	6.03	13.34	2.14	7.17
32x4	7.3	14.37	7.12	9.6
4x4	8.09	14.94	7.22	10.08

Table 2 | Macro FER scores of SongMAE micro ablations. Increasing temporal and frequency resolution produces lower error rates. Lower error rates are better.

We begin with the linear probe, the most direct way to ask whether syllable-relevant information is present in the representations without imposing any assumptions about the geometry of the embedding space [cite cite]. Almost all SongMAE configurations achieve strong linear probe performance - most of the SongMAE micro configurations already outperform the bigger bioacoustic encoders **Table 3**. That a simple linear transformation suffices to produce low classification error rates indicates that the pretraining objective has already organized the embedding space around syllable-relevant structure, and that this organization generalizes to species absent from the pretraining corpus.

We first search over the Voronoi C parameter using the 32 mel bin x 1 time bin configuration. We find that $C = 0.05$ performs best, with a modest gain over $C = 0.1$ and $C = 0.2$, while $C = 0.025$ is noticeably worse. Carrying this value forward, we compare standard MAE random masking against Voronoi masking. As expected, random masking scores worse than the equivalent Voronoi configuration. This gap likely reflects the pretraining objective becoming easier when patches span only 5 ms, since the model can reconstruct masked regions through local interpolation rather than learning meaningful acoustic structure. This is corroborated by the lower C values outperforming the higher C values, as the lower C values correspond to larger contiguous masks. We expect this gap to widen with more data and larger models, as these micro ablations are probably too small to learn features much beyond low-level visual ones.

With respect to patch shape, the best performing SongMAE model is the 16 mel bin x 1 time bin configuration, which is expected since it has the finest temporal resolution together with the finest

frequency resolution. However, the reduction in syllable classification error over the coarser 32 mel bin x 1 time bin model is marginal, likely rendering the modest boost not worth the 2x increase in memory and squared growth in compute. The 128 mel bin x 1 time bin ablation is a notable exception, performing far worse than the other configurations. Because a single 128 mel bin patch spans the entire frequency axis, the model receives no frequency locality within a patch and must encode a full spectral slice into one token, which appears to coarsen the representation enough to degrade syllable classification.

For the coarser configurations, we find that the error rates are higher than those of the finer configurations, where they are especially hurt on Bengalese finch syllable classification. This makes sense, as the Bengalese finch tends to have very fast transitions between syllables (Supplement N). We also find that in our case the coarse 32 x 4 model actually outperforms the 4 x 4 model, indicating that even at matched 20 ms resolution the asymmetric patch shape retains an advantage over the square control. From the micro ablations we carry forward two configurations, both trained at the base preset for all subsequent analysis: 32 x 1, the best fine patch shape, and 32 x 4, the best coarse patch shape.

SongMAE vs others Linear Probes				
Model	Canary	Zebra	Bengalese	Mean
Base 16x4	6.76	13.24	7.11	9.04
Base 32x1	5.22	11.79	1.96	6.32
BirdAVES base	8.12	15.62	8.48	10.74
HuBERT base	14.84	19.94	8.99	14.59

Table 3: Linear probe performance measured with macro FER across SongMAE, BirdAVES, and HuBERT. Scores are averaged across individuals within each species. Patch notation is frequency $\times$ time.

We now compare both carried-forward configurations against BirdAVES and HuBERT at the base preset (Table 3). The coarse 32 x 4 model matches the 20 ms resolution of both baselines, which isolates the effect of architecture from that of temporal resolution, while the fine 32 x 1 model retains the fine temporal patching that distinguishes SongMAE. At matched resolution, 32 x 4 modestly outperforms BirdAVES despite having an order of magnitude fewer parameters. This is consistent with findings in human speech and general audio, where MAE-based architectures have outperformed HuBERT-based approaches [8], and suggests the architectural advantage extends to the bioacoustic domain. The fine 32 x 1 model widens the gap further, outperforming both baselines across all three species.

Taken together, these results suggest that for birdsong, patch resolution along both axes is a dimension of scaling in its own right, one that is largely orthogonal to parameter count. The gains from finer temporal resolution outweigh those from finer frequency resolution, and 32 mel bins x 1 time bin sits at a sweet spot where the temporal resolution is fine enough to resolve rapid syllable transitions while the frequency extent stays coarse enough to keep compute and memory tractable.

5.2. Latent Space Analysis

A well-organized latent space, where acoustically distinct syllables occupy distinct regions, is highly useful for birdsong analysis. Such structure allows for the discovery of units of birdsong [cite cite cite], tracking the change of song over seasons or juvenile development [cite cite], and possibly even individual identification [cite cite]. Prior work has shown that self-supervised models trained on birdsong spectrograms can excel at these tasks due to their structured and organized latent spaces [5, 22, 23]. However, existing approaches have focused entirely on analyzing the latent spaces of birds present in the training set, and have often been tailored to a single species [TweetyBERT, VAE cite].

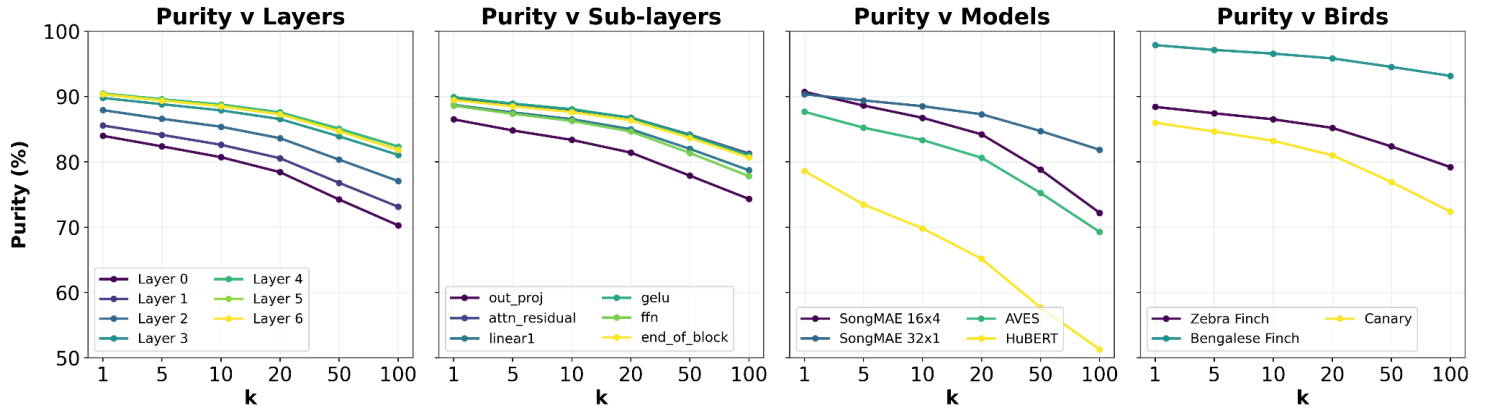


Figure 3: UMAP projections of SongMAE encoder embeddings for canary, Bengalese finch, and zebra finch, colored by ground-truth syllable labels (silence is black). Each point represents a single 5 ms patch embedding from the residual stream of the encoder.

To quantify latent space quality, we use kNN purity rather than the standard approach of dimensionality reduction, clustering, and cluster alignment scores. We choose it for two reasons. First, it is almost parameter-free, reading the structure of the latent space directly through a single interpretable parameter k . Clustering-agreement metrics such as V-measure or ARI, common in the syllable-discovery literature [cite cite cite cite], instead depend on a whole pipeline of dimensionality reduction and clustering, a large space of interacting hyperparameters whose optimal values likely differ across models, confounding cross-model comparison. Second, it is well established in vision as a complement to the linear probe, where frozen features are scored by nearest-neighbor agreement without any fitting [cite; DINO]. It also probes a stricter property: a linear probe only requires that syllables be linearly separable somewhere in the space, whereas purity requires that same-syllable points actually be nearest neighbors under the raw metric, which is exactly the structure that clustering-based annotation and unit discovery depend on. High purity means same-syllable embeddings sit next to one another, so the latent space is locally organized around syllable units with no supervised readout. We report purity for SongMAE, HuBERT, and BirdAVES (Figure 5), and select the layer for our UMAP visualizations by highest purity.

Qualitatively, we find that the UMAP dimensionally reduced SongMAE latent space are highly organized for the three test species. We take the latent space to be the representation encoded in the residual stream, which we then reduce to 2 dimensions for visualization. Because SongMAE operates at 5 ms temporal resolution, we can examine how representations evolve at the timescale

of individual syllables and transitions, a capability unavailable to models operating at coarser resolutions.

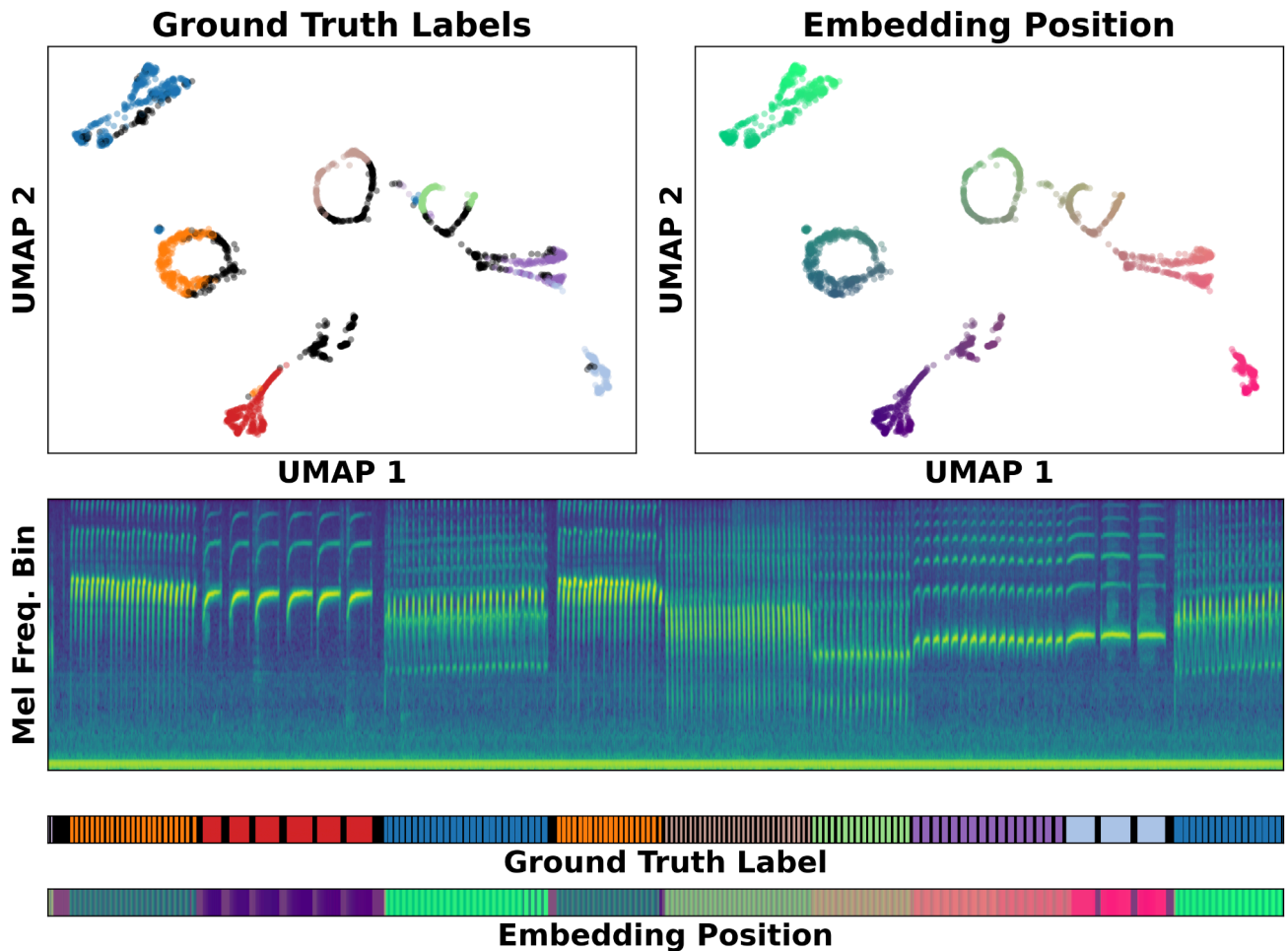


Figure 4: SongMAE embeddings colored by ground-truth labels (left) and UMAP position (right). The figure shows 2,000 time bins (10 s). Color bars link each spectrogram time bin to its label and embedding location (black designates the silence class).

Figure 4 shows the UMAP projection of 2,000 consecutive time bins (10 s) of a canary song, where each point corresponds to a concatenation of patches corresponding to a singular time bin extracted from the residual stream at that 5 ms time step. Despite never encountering canary vocalizations during pretraining, the model's representations clearly segregate syllable types into distinct spatial clusters. Inter-syllable silences (black points) cluster according to their phrasal context: because canaries sing in phrases of repeated syllables, brief silences between repetitions remain embedded near their adjacent syllable cluster rather than forming a single unified silence region.

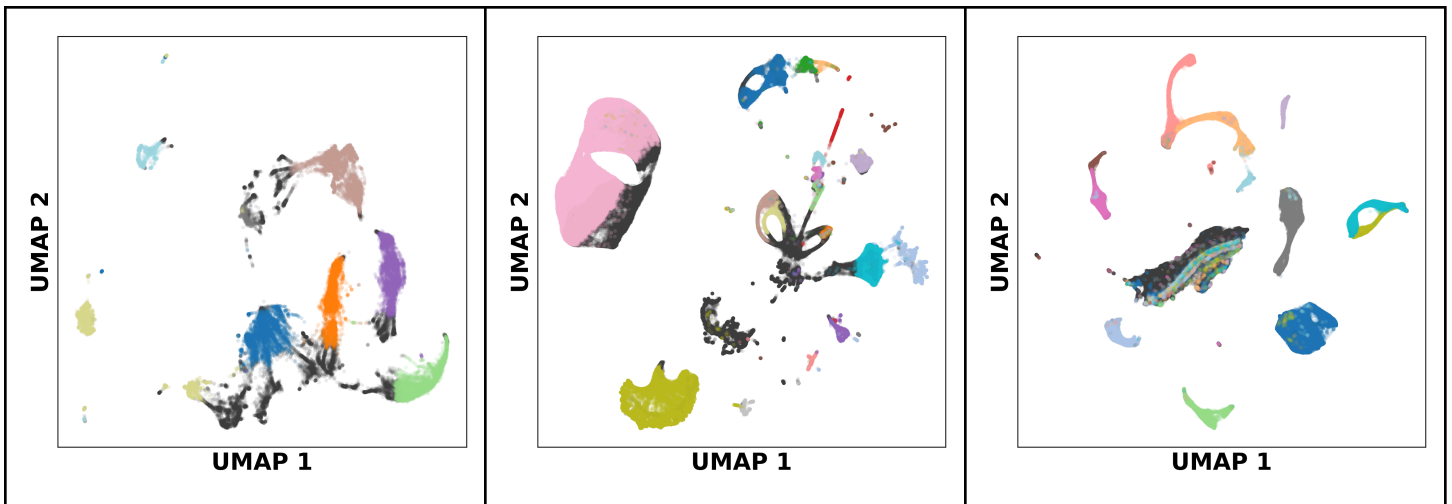


Figure 5: UMAP projections of SongMAE encoder embeddings for canary, Bengalese finch, and zebra finch, colored by ground-truth syllable labels (silence is black). Each point represents a single 5 ms patch embedding from the residual stream of the encoder.

This organization is not limited to canaries or a single song. By projecting many renditions at once, we find that the same syllable type maps consistently to the same region of embedding space, confirming that the clustering in Figure 3 is not an artifact of a single favorable excerpt. Figure 5 shows UMAP projections for the individual bird with the most recorded audio from each species: canary (520 s), Bengalese finch (518 s), and zebra finch (416 s). Zebra finch and Bengalese finch projections similarly show clear spatial segregation of syllable types, demonstrating that SongMAE's representations generalize across species spanning a range of vocal complexity.

5.25 Song look up

5.3. Supervised Fine Tuning

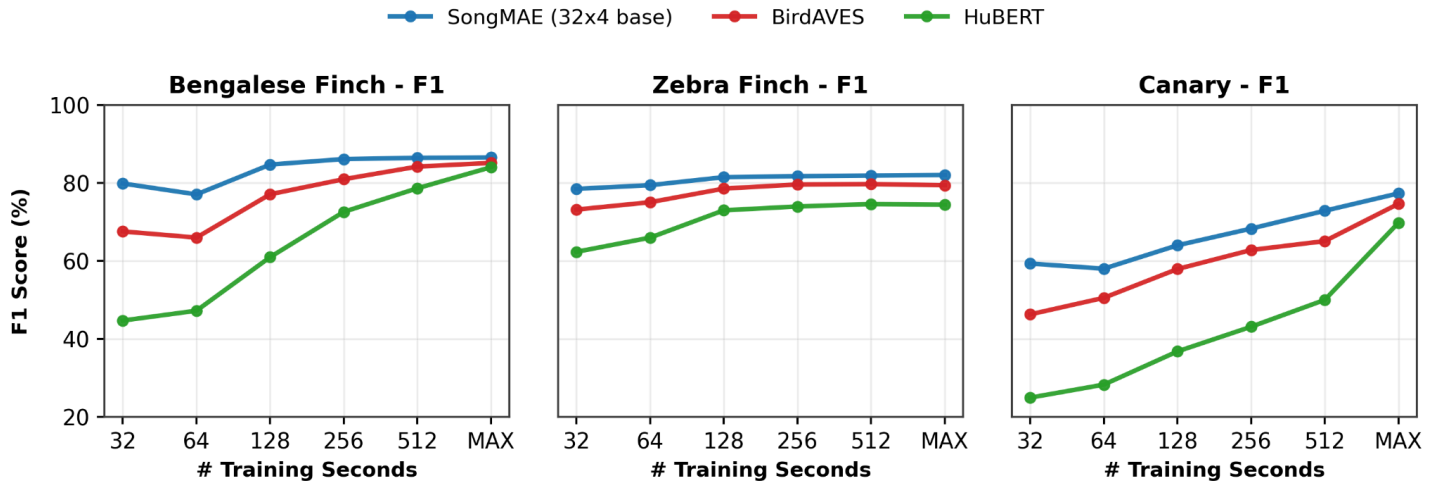


Figure 6: Syllable classification performance (macro F1) across training budgets for Bengalese finch (left), zebra finch (middle), and canary (right). SongMAE is shown in blue, BirdAVES in red, and HuBERT in green. Each line is the mean across birds within a species.

With sufficient labeled data and species-specific post-processing, syllable classification can be solved by relatively simple supervised networks [14, 24]. However, syllable repertoires are individual-specific, meaning any classification network must be trained for each bird from limited labeled examples. Producing such annotations requires domain experts to manually classify individual vocalizations, a process that is time-intensive, costly, and for many species nonexistent. The value of a pretrained encoder, therefore, lies not in its performance given abundant labels, but in its ability to generalize from limited supervision.

To evaluate this, we adopt a few-shot learning paradigm, finetuning each model with $t = \{32, 64, 128, 256, 512, \text{max}\}$ s of labeled audio per individual bird, where max is either the bird's entire training corpus or an upper bound of one hour, whichever is smaller. This setup mirrors realistic deployment, where a researcher may have only a few minutes of annotated recordings for a new individual or species. At every budget, we place separate recordings in the train and test sets and ensure that each syllable type appears in both.

We find that SongMAE outperforms both BirdAVES and HuBERT at all training budgets. This result holds across both fine-grained and coarse-grained patch shapes of SongMAE. The fine-grained SongMAE maintains a large gap in F1 over HuBERT and BirdAVES, while the coarse-grained SongMAE retains only a modest boost and makes more errors overall, likely penalized by its coarser 20 ms resolution straddling syllable boundaries.

For all models, canary requires the most labeled data to reach high F1. This is expected: unlike zebra finch and Bengalese finch, canaries have substantially larger syllable repertoires, so at the minimum budget of 32 s, many syllable types may appear only a handful of times. Furthermore, canary song is unusually complex, with the syntactic arrangement of song units varying from rendition to rendition [cite].

As expected, at least for the SongMAE model whose temporal resolution matches that of HuBERT and BirdAVES, the gap in F1 scores narrows as the training budget increases. This is intuitive: with

fewer labeled examples, the models must lean more heavily on their pretrained representations, but as the budget grows, those representations matter less and architectural inductive biases come to dominate.

6. Discussion

Our results suggest that SongMAE learns generalizable representations of individual song components. Reconstructions from species entirely absent from the pretraining set demonstrate that the model infers large missing fragments of song structure from small contextual clues, linear probes confirm that syllable representations are largely linearly separated, dimensionality reduction of the latent states show that syllables across various species are segregated in different parts of the embedding,

strong representation quality across all patch configurations.

UMAP projections of the encoder’s latent space show that human-annotated syllable types segregate into distinct clusters without any supervision, and this holds across all three test species. When finetuned with limited labels, SongMAE matches or exceeds BirdAVES in low-data regimes despite having an order of magnitude fewer parameters, and the gap widens consistently as more labeled data becomes available, reflecting the advantage of 5 ms temporal resolution.

These findings point toward practical applications in bioacoustic monitoring. The model’s unsupervised syllable clustering could accelerate annotation workflows by providing initial soft labels for species lacking human annotations, reducing one bottleneck in birdsong research. The same representations open possibilities for dialect mapping and population differentiation through the detection of shifts in specific song units or calls [27]. SongMAE’s compact size (~14M parameters) makes it particularly suited for embedded devices in remote ecological monitoring, where computational resources are limited and storing months of raw audio is impractical; the model can operate entirely on-device, producing compressed embeddings for real-time analysis in low-connectivity settings [28].

Several limitations remain. We have evaluated only three species, all recorded in controlled laboratory settings, and generalization to noisier field environments or species with highly variable songs is untested. For such species, more sophisticated approaches to extracting syllable-level structure from the latent space may be required, whether through improved finetuning strategies or analysis methods beyond dimensionality reduction and clustering. SongMAE is also currently undertrained and likely underparameterized; we expect that data augmentation, corpus curation, and longer pretraining will yield substantial improvements. In particular, the pretraining corpus contains long stretches of silence and environmental noise, and filtering with a song activity detector could significantly increase the density of vocal content encountered during training. Our immediate next steps include evaluating additional species for fully unsupervised syllable discovery, scaling model capacity alongside pretraining improvements, and testing on field recordings to assess robustness to real-world acoustic conditions.

7. Conclusion

8. References

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9. Supplements

Before pretraining, we filtered the XCL subset of BirdSet to remove all eBird taxa corresponding to the downstream evaluation species. Specifically, we excluded Island Canary / canary-related entries (`comcan`, `islcan1`, `x01005`), Zebra Finch entries (`zebfin2`, `zebfin3`, `zebfin1`, `chefin1`, `x00906`), Cassin's Vireo entries (`casvir`, `casvir1`, `casvir2`, `y00485`, `y00484`, `solvir1`), American Robin entries (`amerob`, `amerob1`, `amerob2`, `amerob3`, `y00822`), and Bengalese Finch / White-rumped Munia entries (`whrmun`, `whrmun8`, `x01069`). These exclusions include the target species, domestic forms where applicable, identifiable subspecific or regional forms, and ambiguous hybrid, slash, or species-complex categories that could contain the held-out species. We did not remove broader related taxa outside these categories, so the holdout should be interpreted as a species-level exclusion rather than a genus- or family-level exclusion.

5.0 Reconstruction accuracy of unseen species

Fig X:

The first evaluation we conduct is to evaluate the ability of our models to reconstruct spectrograms of songbird species not included in the pretraining corpus. If SongMAE is able to reconstruct species that are not present during the pretraining process, it provides evidence that the model has some degree of generalization. Table 1 shows the MSE error on recordings from 4 sources, aggregated over 1000 spectrogram samples from each source: XCL validation set (pretraining distribution), and three songbird species unseen during pretraining: canaries, zebra finch, and Bengalese finch.

We show the MSE error across different patch and mask configurations. Reconstruction quality cannot be compared between configurations, only within a single one, so we report two within-configuration comparisons: Voronoi versus random masking, and reconstruction quality as a function of training duration.

Numerically, and visually, it's clear that the SongMAE model performs well across all four datasets, and as expected, the voronoi masking produces a substantial boost in reconstruction ability as compared to the random masking.

Supplementary Fig. 1

5.3. Unsupervised Song Detections

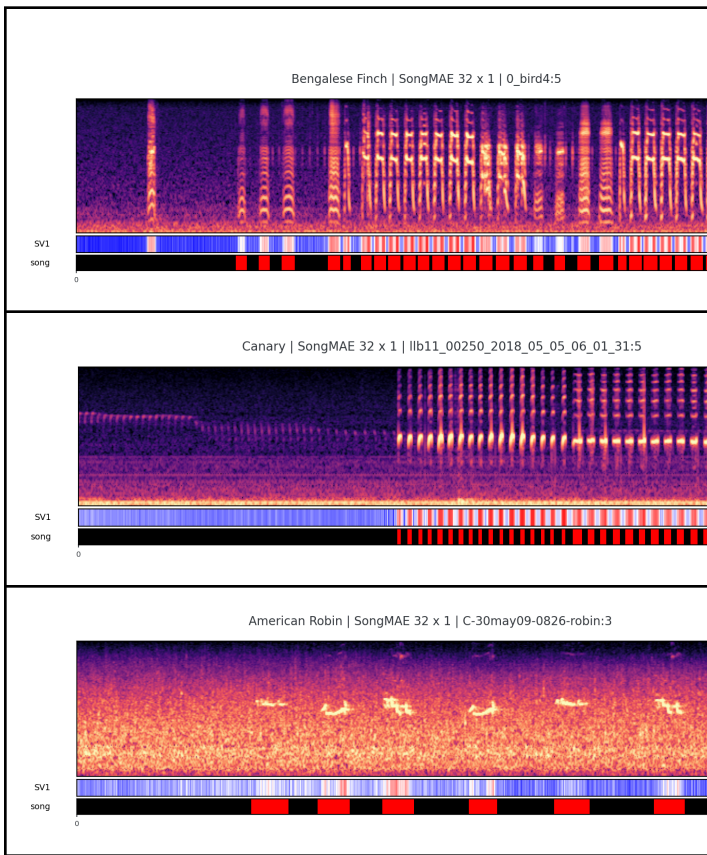


Figure 5:

Frequently recordings of birdsong are unlabeled and unsegmented -- passive audio monitoring and lab recording alike, it is unclear where the units of the song are and where silences and noise are. A naive approach is just to analyze the power of raw audio streams or spectrograms; sequences of audio with lots of energy could indicate a powerful singer. However, this approach fails in certain scenarios, such as when the bird's distance varies from the microphone, or broadband noise appears in the stream of audio -- such as a car or a chainsaw. Supervised approaches at detecting songs / segmenting units have been proliferating in the computational bioacoustics literature, with approaches such as Whisperseg, TweetyNET, DAS, providing researchers a way to generate segmentations and song detections with few manual labels. However, such an approach, while effective for segmenting known species, does not scale to annotating hundreds of unique species. We believe that even an approach that is worse than manual annotation or supervised approach, but better than the naive energy based thresholding could be useful as a first pass annotation or providing soft labels for contrastive neural networks.

SongMAE, unlike HuBERT, can provide a collection of such labels. We found that the first singular vector of the embedding space correlates to "songness" much more than energy of the spectrogram (which we contrast to our method as the pixel intensity at a particular timebin). We calculate this by

finding -- per species of interest - the first singular vector, and then for each recording, computing the dot product of the first singular vector with the embedding vector.

Qualitatively, we find that the dot product value is greater for areas of the spectrograms that are occupied by calls or syllables of a song. Quantitatively, we find that the R^2 is much greater than chance or that of simply energy based calculations.

Although this method is far from perfect, it

